

Influence of intracanal cryotherapy on postendodontic pain and interleukin-6 expression using different irrigation protocols: A randomized clinical trial

Ahmed Emad, Nelly Abdelsalam, Dalia Mukhtar Fayyad

Department of Endodontics, Faculty of Dentistry, Suez Canal University, Egypt

Address for correspondence: Dr. Ahmed Emad, Faculty of Dentistry, Suez Canal University, Port Said, Egypt.

E-mail: ahmedemad_111@hotmail.com

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INTRODUCTION

Apical periodontitis (AP) occurs initially as an inflammatory reaction to microorganisms colonizing the root canal system, then as an infectious disease when bacteria exit the root canal into the periapical tissue. It is similar to any other connective tissue inflammation with the same sequence of inflammatory events including the vasodilatation of blood vessels, the increase in vascular permeability, and the migration of the inflammatory cells such as lymphocytes, macrophages, and neutrophils to the inflamed site.^[1]

The principal immunologically effector cells of AP were macrophages and T helper cells.^[1,2] Release of cell signaling molecules called cytokines helps in communication between cells in immune responses and enhance cell recruitment toward the infected and inflamed sites. Cytokines are present in many forms such as peptides, proteins, and glycoproteins. The development of AP is usually regulated by many cytokines such as interleukin-1 (IL-1) β , IL-6, and tumor necrosis factor.^[3-6]

ILs representing the functional cytokines are the main mediators of inflammatory reactions. Local effects include increased adherence of leukocytes to endothelial walls, lymphocytes stimulation, potentiation of neutrophil, increased production of prostaglandin and proteolytic enzymes, and mediate resorption of bone.^[5]

IL-6 is a primary mediator in the induction and control of the acute phase of inflammation. After an injury, plasma concentrations of IL-6 can be measured in 60 min with a median rise of 4–6 h and can last for up to 10 days. It is considered the most important marker for the degree of tissue damage during surgical procedures. As excessive and prolonged preoperative production of IL 6 during acute phase of inflammation usually results in higher postoperative pain levels.^[7-9]

Postendodontic pain most often occurs during the first 24–48 h following cleaning and shaping and generally recedes in a few hours although it occasionally persists for several days. Pain management during and after root canal treatment is one of the most important aspects of endodontic practice. To control preoperative and postoperative pain, various strategies have been suggested. One of the simplest and effective strategies for pain control in dentistry is “cryotherapy.”^[10]

Employing cryotherapy during root canal treatment has recently gained popularity. Hence, this study was carried out to evaluate the effect of cold temperature of irrigants in reducing postendodontic pain and inflammation.

MATERIALS AND METHODS

The study was given approval by the Ethics Committee, Faculty of Dentistry, Suez Canal University (56/2017) and was registered on clinicaltrials.gov with ID: NCT04324398. Before starting the treatment procedures, all participants were discussed about the nature and objectives of the study, along with obtaining an authored consent.

A total sample size of 48 patients (12 for each group) suffering from symptomatic AP as defined by the American Association of Endodontists was considered to be sufficient to detect this high effect size of $f = 0.5$, a power of 80%, and a 5% significance level (calculation of sample size was done by IBMTM, Sample PowerTM Version 3.0.1 (IBM Corporation, U.S.A.), based on previous studies.^[11,12]

The patients were selected from the Department of Endodontics Outpatient Clinic, Faculty of Dentistry, Suez Canal University. Inclusion criteria were patients aging from 20 to 50 years suffering from symptomatic AP with a preoperative pain score > 7 on the visual analog scale (VAS) scoring.^[13] The exclusion criteria were patients presenting with periapical abscess; patients under immunosuppressive chemotherapy, anti-inflammatory drugs, or antibiotic medications for the past 2 months; medically compromised patients; and pregnant or lactating females.

The selected patients were randomly classified into four groups (12 each) based on irrigation protocols. They were chosen from 48 opaque and tightly sealed envelopes containing the patients' coding. This study was double blinded by the observer and statistician. The only who knew which data belong to which group was the allocator.

Patients' grouping and evaluation methods

Group I (control): irrigation with 5% sodium hypochlorite (NaOCl) at room temperature (RT) from the beginning of cleaning and shaping, between each file. Final rinse was done by 20 mL of 5% NaOCl for 5 min. Group II: irrigation with 5% cold NaOCl (2°C–5°C) from the beginning of cleaning and shaping and between each file. Final rinse was done by 20 mL of 5% cold NaOCl (2°C–5°C) for 5 min. Group III: irrigation with 5% NaOCl at RT from the beginning of cleaning and shaping and between each file. Then, final rinse with 20 mL of 5% cold NaOCl (2°C–5°C) for 5 min. Group IV: irrigation with 5% NaOCl at RT from the beginning of cleaning and shaping and between each file, then final rinse with 20 mL of cold saline (2°C–5°C) for 5 min. Each root canal received a total of 30 mL irrigation using side-vented (30G) needle.

Clinical procedures of root canal treatment

Root canal therapy was conducted in two appointments. At the first appointment, all patients were anaesthetized using one carpule of articaine hydrochloride 40 mg/mL + 1/100,000 epinephrine bitartrate solution (Inibsa Dental, Barcelona, Spain). Cases in which additional anesthesia was needed during access cavity, another carpule of articaine hydrochloride 40 mg/ml + 1/100,000 epinephrine bitartrate solution was administered.

The access cavity was performed under absolute rubber dam isolation using sterile round carbide bur under copious water cooling; then, the initial glide path was obtained using #10 K-file (Micro Méga, Besançon, France).

Coronal flaring was done using SX rotary file (ProTaper, Dentsply Maillefer, Ballaigues, Switzerland), then root canals irrigation with 2 mL 5% NaOCl. The working length at 0.5 was determined using an electronic apex locator (Root ZX Mini; J. Morita Co, Tustin, CA, USA)^[14] and confirmed by periapical radiograph. Biomechanical preparation was carried out by ProTaper Next rotary files (Dentsply Maillefer, Ballaigues, Switzerland) following the instructions of the manufacturer till X3 file.

Evaluation of postendodontic pain

Patients were told about the pain they might experience in the days following treatment and were told to fill a modified VAS with the following criteria: 0 indicated no pain, 1–3 indicated mild pain, 4–6 indicated moderate pain, 7–9 indicated severe tolerable pain, and 10 indicated severe intolerable pain to assess their postoperative pain scores at 12, 24, 48, and 72 h and after 1 week to be submitted at the next appointment, scheduled after 1 week.^[15]

Sample collection for evaluation of gene expression of cytokines

In each group, after obtaining a glide path using #10 K-file (Micro Méga, Besançon, France), sample collection was done before and immediately after root canal cleaning and shaping and then after 1 week for characterization of the messenger RNA (mRNA) expression profile of IL-6. Root canals were dried, followed by inserting three paper points into the root canal to pass 2 mm beyond the apical foramen into the periapical tissues and was hold in place for 1 min to be soaked by the periapical interstitial fluid. 4 mm from the tip of each paper point was cut and dropped into a microcentrifuge tube then stored at -70°C .

The extraction of pure RNA was done using a total RNA purification kit following the protocol of the manufacturer (Thermo Scientific, Fermentas, #K0731).

Complementary DNA (cDNA) synthesis was carried out using RevertAid H minus Reverse Transcriptase to convert RNA into cDNA.

The absorption of ultraviolet (UV) light by the ring structure of purines and pyrimidines for very pure samples (without significant contamination from protein, phenol, free nucleic acids, organic solvents, carbohydrates, etc.) was used to measure the amount of nucleic acids to quantify the concentration of RNA and cDNA to ensure that the purity of the concentrations is adequate to conduct real-time polymerase chain reaction (PCR). The spectrophotometer Q5000 (UV-Vis/USA) was used to ensure accuracy for all the measurements and calculations.^[16]

Real-time polymerase chain reaction

The expression of mRNAs of target genes in periapical interstitial fluid was measured using real-time PCR with SYBR[®] Green (a trade name for cyanine dye produced by Synergy brands Inc. [SYBR]) and an internal reference using β -actin. Amplification of the isolated cDNA was done using 2X Maxima SYBR Green/ROX quantitative PCR Master Mix according to the instructions of the manufacturer (Thermo Scientific, USA, # K0221) and gene-specific primers [Table 1].

The mixture of the final reaction was inserted into a step one plus real-time thermal cycler (Applied Biosystems, Life Technology, USA), and the program of PCR was carried out in 45 cycles under the conditions of PCR: initial denaturation at 95°C for 10 min, denaturation at 95°C for 15 s, annealing at 60°C for 30 s, and extension at 72°C for 30 s. The temperature was increased from 60°C to 95°C at the end of the last cycle to produce a melting curve. The housekeeping gene (β -actin) represented as normalized used to calculate the relative gene expression or the fold change in the target gene. Therefore, the quantities critical threshold (C_t) of the target gene was normalized with quantities C_t of housekeeping gene using the $2^{-\Delta\Delta C_t}$.^[17]

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